KAIZEN MATHS

Lesson starter

Number Bases and Number Sets

Mixed Exam Practice

Name _____

S1

Calculate 45 6 - 44 6 . Give your answer in base 6.

S2

Convert 1110 2 to denary.

S3

State the smallest number set that contains 0.333....

S4

Convert 1100 2 to denary.

Show your working

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NUM-5141-L4 · S

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Find the digit x if x3 8 = 27 10 .

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Calculate 21 3 + 100 3 . Give your answer in base 3.

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Find the digit x if x3 4 = 11 10 .

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Teacher answers & working

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S1

Calculate 45 6 - 44 6 . Give your answer in base 6.

Answer:

1_6

Working:

Convert each number to denary first.

$45_6 = 29_{10}$

$44_6 = 28_{10}$

$29 - 28 = 1$

Convert 1_{10} back to base 6: 1_6.

S2

Convert 1110 2 to denary.

Answer:

14_{10}

Working:

Use powers of the base as place values.

$1110_2 = 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$

$1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 = 1 \times 8 + 1 \times 4 + 1 \times 2 + 0 \times 1$

$1 \times 8 + 1 \times 4 + 1 \times 2 + 0 \times 1 = 14$

So $1110_2 = 14_{10}$.

S3

State the smallest number set that contains 0.333....

Answer:

$\mathbb{Q}$

Working:

Start with the most specific description of 0.333....

0.333... is a recurring decimal, so it is rational.

The smallest suitable set is $\mathbb{Q}$: rational number.

Teacher answers & working

Number Bases and Number Sets · Mixed Exam Practice

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S4

Convert 1100 2 to denary.

Answer:

12_{10}

Working:

Use powers of the base as place values.

$1100_2 = 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0$

$1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0 = 1 \times 8 + 1 \times 4 + 0 \times 2 + 0 \times 1$

$1 \times 8 + 1 \times 4 + 0 \times 2 + 0 \times 1 = 12$

So $1100_2 = 12_{10}$.

E1

Find the digit x if x3 8 = 27 10 .

Answer:

3

Working:

Expand using place value in base 8.

$x3_8 = 8x + 3$

$8x + 3 = 27$

$8x = 24$

$x = 3$

Check: 3 is a valid base-8 digit because 3 < 8.

E2

Calculate 21 3 + 100 3 . Give your answer in base 3.

Answer:

121_3

Working:

Convert each number to denary first.

$21_3 = 7_{10}$

$100_3 = 9_{10}$

$7 + 9 = 16$

Convert 16_{10} back to base 3: 121_3.

Teacher answers & working

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E3

Find the digit x if x3 4 = 11 10 .

Answer:

2

Working:

Expand using place value in base 4.

$x3_4 = 4x + 3$

$4x + 3 = 11$

$4x = 8$

$x = 2$

Check: 2 is a valid base-4 digit because 2 < 4.